

FIELD WORK DETAILS

Problem Statement ID : SIH26036

Problem Statement Title : Development of an Online Verification System for Weighing and Measuring Instruments.

Theme : Miscellaneous

PS Category : SOFTWARE

Team Name : DEXAT



NAME: Government Approved Test Centre , OMR.

ROLE : Vikram & Rajesh – Lab Technicians

Problems Faced by Government Approved Test Centres (GATCs)

- **Manual Compliance Audits:** Relies on paper records for 5-year compliance histories, causing heavy storage and audit friction.
- **Inconsistent Verification Data:** Inputs testing results manually, creating frequent record mismatches with central government databases.
- **Tamper-Prone Stamping:** Issues physical paper certificates that are vulnerable to counterfeiting and fraudulent duplication.
- **Verification Backlogs:** Manages trader scheduling manually, resulting in long testing queues and operational delays.



NAME: Legal Metrology Officers (LMOs), Chennai.

ROLE : Sudha - Labour Department office staff

Also connect with the State LMO via phone call.

Problems Faced by Legal Metrology Officers (LMOs)

- **Manual Paperwork:** Wastes time filling paper registers, hand-writing notices, and issuing physical certificates.
- **Random Inspections:** Conducts unguided field visits without data on high-risk zones or uncalibrated scales.
- **Certificate Fraud:** Struggles to detect illegal scale tampering and counterfeit paper calibration stamps on-site.
- **Manual Calculations:** Computes permissible device error margins and tolerance limits manually during inspections.
- **Fee Collection:** Delays payment processing on-site due to reliance on physical receipts a manual reconciliation.

Main Problems Faced by Legal Metrology Department

- **Revenue Leakage:** Loses significant state revenue due to untracked commercial scales, expired licenses, and missed renewal fees.
- **Chennai** inefficient, random routine checks instead of targeting non-compliant commercial hubs.
- **Systemic Corruption:** Faces fake paper certificates, ghost audits, and unregistered measuring devices due to a lack of digital verification.
- **Information Silos:** Operates with fragmented paper records across districts, preventing real-time cross-district monitoring and decision-making.
- **Consumer Protection Gaps:** Lacks public-facing tools for citizens to verify device accuracy, leading to unpunished retail market short-weighting.
- **High Administrative Overhead:** Incurs massive paper, printing, and storage costs to manage millions of physical license files across regional offices.
- **Unmonitored E-Commerce Scaling:** Struggles to regulate fast-growing automated logistics, digital flow meters, and modern electronic weighbridges.